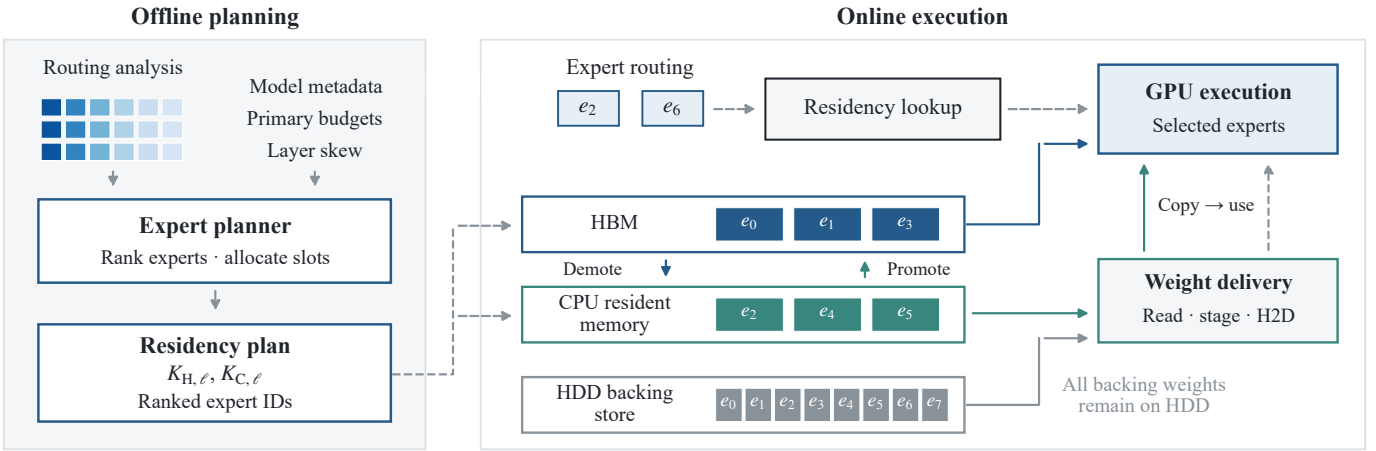

Offline planning
Online execution
Routing analysis
Model metadata
Primary budgets
Layer skew
Expert planner
Rank experts · allocate slots
Residency plan
$K_{H,\ell}$, $K_{C,\ell}$
Ranked expert IDs
Expert routing
e_2
e_6
Residency lookup
GPU execution
Selected experts
HBM
e_0
e_1
e_3
Copy → use
Demote
Promote
CPU resident memory
e_2
e_4
e_5
Weight delivery
Read · stage · H2D
HDD backing store
e_0 e_1 e_2 e_3 e_4 e_5 e_6 e_7
All backing weights remain on HDD
Dashed: control / plan metadata
Solid: weight movement